

BRAMON, Sweden

WMO: 023680

Lat: 62.220N

Lon: 17.739E

Elev: 59

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.4	10.4	2.2	6.2	8.0	5.9	7.5	12.0	37.2	28.5	34.7	27.3	11.1	320	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	71.8	64.5	69.3	63.0	67.0	61.4	66.4	70.0	64.3	67.8	62.4	65.6	9.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.1	68.7	62.8	85.7	66.6	60.9	80.0	64.4	31.1	70.2	29.5	67.9	28.1	65.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	28.5	25.1	2.9	77.4	6.6	4.1	-1.8	80.4	-5.7	82.8	-9.4	85.1	-14.2	88.1		
(5)			1.5	66.7	6.7	2.3	-3.3	68.4	-7.2	69.8	-11.0	71.1	-15.9	72.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	27.7	27.3	31.4	37.9	46.0	54.7	61.1	60.1	53.0	43.5	36.0	30.8		
	DBStd	13.35	7.97	8.73	7.26	5.12	5.17	4.72	4.33	4.08	4.62	5.10	5.86	7.62		
	HDD50	3673	693	636	576	364	147	10	0	0	24	208	421	594		
	HDD65	8225	1158	1056	1041	813	590	310	140	161	361	666	870	1059		
	CDD50	950	0	0	0	1	22	151	343	313	113	7	0	0		
	CDD65	27	0	0	0	0	0	1	17	9	0	0	0	0		
	CDH74	30	0	0	0	0	1	1	24	4	0	0	0	0		
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0		
	WSAvg	11.6	13.5	13.0	12.2	10.9	9.9	10.6	10.1	10.4	10.9	12.6	13.0	12.7		
	PrecAvg	21.4	1.3	0.9	1.0	1.3	1.6	1.9	2.2	2.8	2.1	2.4	2.2	1.6		
	PrecMax	33.1	2.9	1.7	2.4	3.1	3.0	3.1	6.5	5.3	4.9	6.1	6.3	3.3		
(15) Precipitation	PrecMin	15.7	0.1	0.3	0.2	0.2	0.2	0.2	0.4	0.6	0.6	0.2	0.6	0.1		
	PrecStd	4.1	0.7	0.4	0.6	0.9	0.8	0.8	1.4	1.3	1.0	1.7	1.4	0.8		
	DB	44.3	46.6	51.3	57.6	64.4	71.3	76.6	73.8	67.2	57.3	51.5	46.9			
	MCWB	39.6	40.1	43.1	48.2	52.2	58.5	65.2	66.1	58.5	51.8	46.2	43.1			
	2% DB	41.2	43.2	47.1	52.8	59.8	67.5	72.9	70.9	63.4	54.0	47.3	44.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	37.4	38.1	40.3	44.8	51.3	57.8	65.1	65.0	57.5	50.6	44.4	40.8			
	5% DB	39.2	40.4	44.1	48.9	56.7	64.4	70.3	68.4	61.1	52.2	44.9	41.9			
	MCWB	35.9	36.3	38.5	42.3	49.1	56.2	64.3	63.1	56.3	49.4	43.0	39.0			
	10% DB	37.3	37.3	41.1	45.7	53.6	61.8	67.9	66.3	59.2	50.5	43.2	40.1			
	MCWB	34.7	34.1	36.3	40.4	47.4	55.1	62.7	61.7	55.2	47.8	41.6	37.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	40.3	40.6	43.8	48.4	54.7	61.2	69.5	69.0	60.7	53.3	46.8	44.0			
	MCDB	43.4	45.2	50.3	56.5	61.4	67.0	72.5	71.7	64.0	54.8	49.4	45.8			
	2% WB	38.1	38.4	41.0	45.3	52.0	59.1	67.1	66.3	58.9	51.7	44.9	41.3			
	MCDB	40.7	42.5	46.2	51.7	58.4	65.2	71.0	69.5	61.6	53.4	46.7	43.3			
	5% WB	36.4	36.6	38.7	43.0	49.9	57.6	65.1	64.4	57.5	50.1	43.5	39.6			
(33) Mean Daily Temperature Range	MCDB	38.6	39.6	43.4	48.1	55.5	62.5	69.0	67.3	60.1	51.9	44.7	41.4			
	10% WB	34.9	34.7	36.6	41.0	48.0	56.2	63.1	62.6	56.1	48.1	42.1	38.0			
	MCDB	36.9	36.8	40.5	45.1	52.6	60.5	66.9	65.1	58.6	50.1	43.1	39.7			
	5% DB	8.3	9.8	12.6	14.7	14.0	12.5	10.7	9.7	10.1	8.1	7.2	8.0			
	MCDBR	6.8	7.4	8.2	8.7	7.9	6.5	6.0	6.1	6.5	6.3	6.3	6.7			
(40) Clear-Sky Solar Irradiance	5% WB	7.4	9.1	11.8	14.1	12.8	10.4	8.4	8.0	8.7	6.9	6.4	7.2			
	MCWBR	6.4	7.3	8.0	8.8	7.9	6.2	5.7	5.6	6.5	6.2	6.0	6.5			
	taub	0.244	0.299	0.319	0.356	0.356	0.351	0.370	0.362	0.330	0.316	0.262	0.344			
	taud	2.140	2.268	2.364	2.378	2.443	2.498	2.464	2.478	2.550	2.442	2.190	2.598			
(42) All-Sky Solar Radiation	Ebn at Noon	140	209	252	264	274	278	268	260	250	205	130	87			
	Edh at Noon	10	19	25	31	31	30	30	28	22	16	10	6			
(44) All-Sky Solar Radiation	RadAvg	63	257	731	1211	1579	1811	1665	1264	787	342	94	35			
	RadStd	9	46	83	107	158	114	170	104	92	60	20	3			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-189	N/A	N/A	N/A

Nomenclature: See separate page